

A Practical Guide to Residual connection neural network

TOPIC -- RESIDUAL CONNECTION NEURAL NETWORK

$$a^{(l)} = \sigma(W^{(l)}a^{(l-1)} + b^{(l)})$$

-- 01 --

This formulation suggests that the gradient computation of a shallower layer, $\frac{\partial E}{\partial x_{\ell}}$, always has a later term $\frac{\partial E}{\partial x_L}$ that is directly added. Even if the gradients of the $F(x_i)$ terms are small, the total gradient $\frac{\partial E}{\partial x_{\ell}}$ resists vanishing due to the added term $\frac{\partial E}{\partial x_L}$.

-- 02 --

where L is the index of a residual block and ℓ is the index of some earlier block. This formulation suggests that there is always a signal that is directly sent from a shallower block ℓ to a deeper block L .

-- 03 --

$$\begin{aligned} \frac{\partial E}{\partial x_{\ell}} &= \frac{\partial E}{\partial x_L} \frac{\partial x_L}{\partial x_{\ell}} = \frac{\partial E}{\partial x_L} \left(1 + \frac{\partial x_L}{\partial x_{\ell}} \sum_{i=\ell}^{L-1} F'(x_i) \right) \\ &= \frac{\partial E}{\partial x_L} + \frac{\partial E}{\partial x_L} \frac{\partial x_L}{\partial x_{\ell}} \sum_{i=\ell}^{L-1} F'(x_i) \end{aligned}$$

-- 04 --

where P is typically a linear projection, defined by $P(x) = Mx$ where M is a $m \times n$ matrix. The matrix is trained via backpropagation, as is any other parameter of the model.

-- 05 --

Subsequent work Wide Residual Network (2016) found that using more channels and fewer layers than the original ResNet improves performance and GPU-computational efficiency, and that a block with two 3×3 convolutions is superior to other configurations of convolution blocks. DenseNet (2016) connects the output of each layer to the input to each subsequent layer:

-- 06 --

Applications Originally, ResNet was designed for computer vision. All transformer architectures include residual connections. Indeed, very deep transformers cannot be trained without them. The original ResNet paper made no claim on being inspired by biological systems. However, later research has related ResNet to biologically-plausible algorithms. A study published in Science in 2023 disclosed the complete connectome of an insect brain (specifically that of a fruit fly larva). This study discovered "multilayer shortcuts" that resemble the skip connections in artificial neural networks, including ResNets.

-- 07 --

Bottleneck block A bottleneck block consists of three sequential convolutional layers and a residual connection. The first layer in this block is a 1×1 convolution for dimension reduction (e.g., to 1/2 of the input dimension); the second layer performs a 3×3 convolution; the last layer is another 1×1 convolution for dimension restoration. The models of ResNet-50, ResNet-101, and ResNet-152 are all based on bottleneck blocks.

-- 08 --

where ϕ can be any activation (e.g. ReLU) or normalization (e.g. LayerNorm) operation. This design reduces the number of non-identity mappings between residual blocks, and allows an identity mapping directly from the input to the output. This design was used to train models with 200 to over 1000 layers, and was found to consistently outperform variants where the residual path is not an identity function. The pre-activation ResNet with 200 layers took 3 weeks to train for ImageNet on 8 GPUs in 2016. Since GPT-2, transformer blocks have been mostly implemented as pre-activation blocks. This is often referred to as "pre-normalization" in the literature of transformer models.

-- 09 --

Basic block A basic block is the simplest building block studied in the original ResNet. This block consists of two sequential 3x3 convolutional layers and a residual connection. The input and output dimensions of both layers are equal.

-- 10 --

A residual neural network (also referred to as a residual network or ResNet) is a deep learning architecture in which the layers learn residual functions with reference to the layer inputs. It was developed in 2015 for image recognition, and won the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) of that year. As a point of terminology, "residual connection" refers to the specific architectural motif of $x \mapsto f(x) + x$, where f is an arbitrary neural network module. The motif had been used previously (see §History for details). However, the publication of ResNet made it widely popular for feedforward networks, appearing in neural networks that are seemingly unrelated to ResNet. The residual connection stabilizes the training and convergence of deep neural networks with hundreds of layers, and is a common motif in deep neural networks, such as transformer models (e.g., BERT, and GPT models such as ChatGPT), the AlphaGo Zero system, the AlphaStar system, and the AlphaFold system.

-- 11 --

Stochastic depth is a regularization method that randomly drops a subset of layers and lets the signal propagate through the identity skip connections. Also known as DropPath, this regularizes training for deep models, such as vision transformers. ResNeXt (2017) combines the Inception module with ResNet. Squeeze-and-Excitation Networks (2018) added squeeze-and-excitation (SE) modules to ResNet. An SE module is applied after a convolution, and takes a tensor of shape $R \times H \times W \times C$ (height, width, channels) as input. Each channel is averaged, resulting in a vector of shape $R \times C$. This is then passed through a multilayer perceptron (with an architecture such as linear-ReLU-linear-sigmoid) before it is multiplied with the original tensor. It won the ILSVRC in 2017.

-- 12 --

During the late 1980s, "skip-layer" connections were sometimes used in neural networks. Examples include: Lang and Witbrock (1988) trained a fully connected feedforward network where each layer skip-connects to all subsequent layers, like the later DenseNet (2016). In this work, the residual connection was the form $x \mapsto F(x) + P(x)$, where P is a randomly-initialized projection connection. They termed it a "short-cut connection". An early neural language model used residual connections and named them "direct connections".